Lattice Gauge Theory and Algebraic Quantum Field Theory

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Part I

Constructive Quantum Field Theory

1 Rigorous Construction of Quantum Mechanics

In order to define a quantum field theory rigorously, we need to first study quantum mechanics in the cases of finite dimensional spaces of particle number. When studying quantum mechanics, the goal to obtain is a wave function provided a Hamiltonian or an action. To this end, however, there is natural structure that arises including the Hilbert Space one works over. We seek to have a generalized process for all of this. We introduce a couple of theorems to motivate the process.

Theorem 1.1 (Stone's Theorem). Given a strongly continuous one parameter group of unitary operators $(U(t))_{t\in\mathbb{R}}$ on a Hilbert Space $\mathcal{H}$, define

$$\mathcal{D} := \left\{ x \in \mathcal{H} : \lim_{t\to 0} \frac{U(t)x - x}{-it} \text{ exists } \right\}$$

then $\mathcal{D}$ is a dense subspace on $\mathcal{H}$. Thus, defining $H : \mathcal{D} \to \mathcal{H}$ via the limit operation above, we have that H is self-adjoint. In addition, the converse of obtaining $(U(t))_{t\in\mathbb{R}}$ from a given densely defined operator H, there is a unique $\mathcal{D}$ such that the above is satisfied.

Therefore, we call such H the **Hamiltonian Operator** which also satisfies Schrodinger's Equation:

$$i\frac{d}{dt}x(t) = Hx(t)$$

for $x(t) = U(t)x$.

Theorem 1.2 (Hille-Yosida Theorem). Given $(P(t))_{t\geq 0}$ and $\mathcal{H}$ as above, then

$$\mathcal{D} := \left\{ x \in \mathcal{H} : \lim_{t\downarrow 0} \frac{P(t)x - x}{-t} \text{ exists } \right\}$$

is dense in $\mathcal{H}$. Thus, defining $H : \mathcal{D} \to \mathcal{H}$ vai the limit operation above, we have that H is self-adjoint.

Notice the similarity between the two theorems. In particular, quantum mechanics uses both of these theorems to relate a proper notion of a **Wick Rotation** to take the group of operators of the form $P(t) = e^{-tH}$, rotation by "i" from $t \mapsto it$ in the complex plane to get the unitary operators $U(t) = e^{-itH}$. The probabalistic approach is as follows.

We need to first construct a Euclidean Path Integral so that we have a well-defined action to vary in a quantum system, this is done by Wick rotating an action S to S_E. Thus, this gives a valid probability measure ν proportional to $e^{-S_E(\gamma)}$ with respect to some notion of a Lebesgue measure. One can therefore construct a Stochastic Process $X = \{X_t\}_{t\in\mathbb{R}}$ which is χ-valued and has law ν stationary and reversible. In particular, X_t has law μ for all t with probability density f reversible with respect to some λ. Defining $Q(t)\phi(x) := \mathbb{E}(\phi(X_t)|X_0 = x)$, this gives

us the definition for our semigroup of operators $(Q(t))_{t\geq 0}$ for linear operators on $\mathcal{H}$. Thus, using the process outlined with Hille-Yosida and Stone's theorem, we generate a Hamiltonian for our quantum theory along with out Hilbert Space.

2 Rigorous Construction of a Quantum Field Theory

Constructing a quantum field requires upgrading our operators to act at every point in space. Such an example of a classical field satisfies the Klein-Gordon equation of motion:

$$\frac{\partial^2 \phi}{\partial t^2} = \Delta\phi - m^2\phi$$

with action

$$S(\phi) = \frac{1}{2}\int_{\mathbb{R}^4} ((\partial_t \phi(t,x))^2 - ||\nabla\phi(t,x)||^2 - m^2\phi(t,x)^2)\ dtdx$$

so that the Euler-Lagrange equation yields the Klein-Gordon equation. Recall that in quantum mechanics, given a path $\gamma(t)$, we upgrade this to a wavefunction at time t $\psi_t(\cdot)$ where $|\psi_t(x)|^2$ is proportional to the probability density function. Analogously, when we are given $\phi(t,\cdot)$, we want to find $\psi_t(\cdot)$ such that $|\psi_t(\cdot)|^2$ proportional to our desired probability density function. First, we need to find a Schwartzian space so that the probability density of $\phi(t,\cdot)$ at $\xi \in \mathcal{S}$ with respect to some Lebesgue measure is proportional to $|\psi_t(\xi)|^2$. It is suggestive, therefore, to define

$$\psi_t(\xi) = \frac{1}{Z(t)}\int_{\phi:\phi(0,\cdot)=\xi(\cdot)} \psi_0(\phi(t,\cdot))e^{iS(\phi)}\mathcal{D}\phi$$

with $\mathcal{D}\phi$ being a measure over $\{\phi : \phi(s,\cdot) \in \mathcal{S}\ \forall\ 0 \le s \le t\}$.

Let $G_m(x,y)$ be the integral kernel of $-\Delta + m^2 I$, and we can view $\{\phi(t,\cdot)\}_{t\in\mathbb{R}}$ as a stochastic process of fields over time. Define

$$\phi_t(f) = \int_{\mathbb{R}^{d-1}} \phi(t,x)f(x)\ dx$$

so that

$$cov(\phi_s(f), \phi_t(g)) = \int_{\mathbb{R}^{d-1}}\int_{\mathbb{R}^{d-1}} f(x)g(y)G_m(s-t, x-y)\ dydx$$

Now, consider the Feynman propagator

$$\Delta_F(x-y) := i\langle\Omega|\mathcal{T}\{\phi(x)\phi(y)\}|\Omega\rangle$$

which can be viewed broken down in spacetime as

$$\Delta_F(t,x) = \frac{i}{(2\pi)^3}\int_{\mathbb{R}^3} \frac{e^{-i|t|\sqrt{||p||^2+m^2}+ip\cdot x}}{2\sqrt{||p||^2+m^2}}\ dp$$

or instead a bilinear form such that

$$\Delta_F(t,x) = \frac{i}{(2\pi)^3}\int_{\mathbb{R}^3}\int_{\mathbb{R}^3}\int_{\mathbb{R}^3} \frac{e^{-i|t|\sqrt{||p||^2+m^2}}\hat{f}(p)\overline{\hat{g}(p)}}{2\sqrt{||p||^2+m^2}}\ dp.$$

Theorem 2.1. For all $f, g \in \mathcal{S}(\mathbb{R}^3, \mathbb{R})$ and $-\infty < s \leq t < \infty$, we have that

$$\langle \Omega | \phi_s(g)\phi_t(f) | \Omega \rangle = -i\Delta_F(t-s, f, g).$$

The **Wightman functions** are formally defined as $W(x_1, ..., x_n) := \langle \Omega | \mathcal{T}\{\phi(x_1)\phi(x_2)...\phi(x_n)\} | \Omega \rangle$ which rigorously on $(\mathcal{S}(\mathbb{R}^4, \mathbb{C}))^n$ is defined as

$$W(f_1, ..., f_n) := \int_{\mathbb{R}} ... \int_{\mathbb{R}} \langle \Omega | \mathcal{T}\{\phi_{t_1}(f(t, \cdot))...\phi_{t_n}(f_n(t_n, \cdot))\} | \Omega \rangle \ dt_1...dt_n.$$

3 With Particles

In this section, we need to consider Lorentz transformations of the form

$$\eta = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

so that under the following inner product, it becomes an isometry:

$$(x, y) := x_0 y_0 - x_1 y_1 - x_2 y_2 - x_3 y_3$$

of which the space of Lorentz transformations form the Lie group $O(1, 3)$.

Definition 3.1. The restricted Lorentz group $SO^+(1, 3)$ or the orthochronous Lorentz group satisfies $L \in O(1, 3)$ such that $\det(L) = 1$ and $\text{sgn}(L_{00}) = 1$.

Consider mass shells a manifold of the form

$$\chi_m := \{p \in \mathbb{R}^4 : p^2 = m^2, p_0 \geq 0\}$$

We wish to integrate over this mass shell. Consider the project map $\pi : \chi_m \to \mathbb{R}^3$ via

$$\pi(p_0, p_1, p_2, p_3) = (p_1, p_2, p_3).$$

Given a Borel set (which is constructed from the topology of $\mathbb{R}^4$ restricted to χ_m!), we can define

$$\lambda_m(A) := \frac{1}{(2\pi)^3} \int_{\pi(A)} \frac{dq}{2\omega_q}$$

where $\omega_q := \sqrt{||q||^2 + m^2}$ for $q \in \mathbb{R}^3$. For a measurable function $f : \chi_m \to \mathbb{R}$ we also have that

$$\int_{\chi_m} f(p) \ d\lambda_m(p) = \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{f(\omega_q, q)}{2\omega_q} \ dq.$$

3.1 Bosonic Fock Space

Note that the above construction gives a Hilbert space for momentum states of bosons with mass m, namely $L^2(\chi_m, \lambda_m)$. For systems of n-particles, however, given a state f of the system, we want it to be symmetric amongst the particles. That is, we want elements of the Hilbert space $\mathcal{B}_n := L^2_{\text{sym}}(\chi_m^n, \lambda_m^n)$. For arbitrary amount of bosons with indeterminate momenta, we generalize the space to be

$$\mathcal{B} := \widehat{\bigoplus}_{n\geq 0} \mathcal{B}_n.$$

Now, let us reconsider $\mathcal{A} = \mathbb{R}^{\mathcal{S}(\mathbb{R}^{d-1},\mathbb{R})}$, the space of functions from $\mathcal{S}(\mathbb{R}^{d-1}, \mathbb{R})$ to $\mathbb{R}$ and $\mathcal{C}$ denote the cylinder σ-algebra on $\mathcal{A}$. Define the Hilbert space $\mathcal{H} = L^2(\mathcal{A}, \mathcal{C}, \mu)$. Define $\mathcal{H}$ the set of constant random variables. For $n \geq 1$, we define $\mathcal{H}_n$ the closure of the span of Wick ordered monomials of degree n in $\{\psi(f) : f \in \mathcal{S}(\chi_m, \mathbb{C})\}$. Now, define a map $\tau : \oplus_{n=1}^{\infty} \mathcal{H}_n \to \mathcal{B}$ by

$$\tau(:\psi(f_1)...\psi(f_n):)(p_1, ..., p_n) := \frac{1}{\sqrt{n!}} \sum_{\sigma\in S_n} f_1(p_{\sigma(1)})...f_n(p_{\sigma(n)})$$

extended via linearity. In particular, this is an isomorphism between $\mathcal{H}$ and $\mathcal{B}$.

Remark 3.2. The main takeaway is that we developed the map τ to show the pairing between a free quantum field and a system of indeterminate number of bosons. However, we want to also consider the dynamics of an evolving free quantum field, or in particular a system of bosons. Consider the following theorem.

Theorem 3.3. For all $n \geq 1$ and $f_1, ..., f_n \in \mathcal{S}(\chi_m, \mathbb{C})$. In the Fock space representation of a free quantum field such that

$$\psi(p_1, ..., p_n) = \frac{1}{\sqrt{n!}} \sum_{\sigma\in S_n} f_1(p_{\sigma(1)})...f_n(p_{\sigma(n)}) \in \mathcal{B}_n$$

then we can evolve the state as

$$\begin{aligned} U(t)\psi(p_1, ..., p_n) &= \frac{1}{\sqrt{n!}} \sum_{\sigma\in S_n} (e^{-itp_{\sigma(1),0}} f_1(p_{\sigma(1)}))...(e^{-itp_{\sigma(n),0}} f_n(p_{\sigma(n)})) \\ &= e^{-it\sum_{j=1}^n p_{j,0}} \psi(p_1, ..., p_n) \end{aligned}$$

where the operator extends via linearity and continuity.

The proof involves utilizing the free quantum field, with $F, G \in \mathcal{H} = L^2(\mathcal{A}, \mathcal{C}, \mu)$ for μ the law of the time slice ϕ_0 of a Euclidean field ϕ and ψ_0 a random field using ϕ_0. The states are such that

$$F(\phi_0) :=: \psi_0(f_1)...\psi_0(f_n) :, G(\phi_0) :=: \psi_0(g_1)...\psi_0(g_n) :$$

Define $(P(t))_{t\geq 0}$ the semigroup for the Euclidean free field, then utilizing Wick's thoerem and some Fourier analysis we wish to simpify

$$||U(t)F - \overline{G}||_{\mathcal{H}}^2 = \mathbb{E}(|F(\phi_0)|^2) + \mathbb{E}(|G(\phi_0)|^2) - 2\,\mathrm{Re}(\mathbb{E}(G(\phi_0)U(t)F(\phi_0)))$$

where given $f_1, ..., f_n$, we choose $g_1, ..., g_n$ as

$$g_j(p) := \overline{e^{-itp_0} f_j(p)} = e^{itp_0}\overline{f_j(p)}$$

where for $(U(t))_{t\in\mathbb{R}}$, the time evolution group for the free quantum field, we have that

$$\mathbb{E}(G(\phi_0)U(t)F(\phi_0)) = \sum_{\sigma\in S_n}\prod_{j=1}^{n}\int_{\chi_m} e^{-itp_0} g_j(p) f_{\sigma(j)}(\tilde{p})\; d\lambda_m(p)$$

and simplifying the expression gives that $U(t)F = \overline{G}$.

4 Yang-Mills Theory

Yang-Mills Theory is the key language that utilizes gauge theory, putting three of the four fundamental forces unified in the same theory. There will be a heavy use of differential geometry terminology used, as the goal will be to consider actions on manifolds M of dimension n equipped with a Riemannian (or Pseudo-Riemannian) metric g. Let $P = M \times G$ be a trivial principle bundle, where G is a matrix Lie group and $\mathfrak{g}$ the corresponding Lie Algebra of G. A will be the principle G-connection on P, a $\mathfrak{g}$-valued 1-form that is usually called the "gauge field." Denote F as the curvature form, so that $F \wedge *F$ is a $\mathfrak{g}$-valued n-form. Thus, on a chart (U, ϕ), we can write

$$F \wedge *F = H \ dx_1 \wedge ... \wedge dx_n$$

for $H : U \to \mathfrak{g}$ smooth. Thus, we denote

$$\text{tr}(F \wedge *F) = \text{tr}(H) \ dx_1 \wedge ... \wedge dx_n.$$

Therefore, we define our **Yang-Mills Action** as

$$S(A) := -\frac{1}{2g_0^2} \int_M \text{tr}(F \wedge *F)$$

with g_0 the bare coupling constant. In abelian gauge fields, we can rescale coordinates and set g_0 to unity, but this will be a nontrivial coefficient used in nonabelian gauge fields. note that in coordinates, we can rewrite

$$F \wedge *F = \langle F, F \rangle \omega$$

where ω is the volume form of M, and $\langle F, F \rangle = F_{ij} F^{ij}$ with $F^{ij} = \sum_{k,l} g^{ik} g^{jl} F_{kl}$. As an example if we have the standard metric on $M = \mathbb{R}^n$, we get that

$$S(A) = -\frac{1}{4g_0^2} \int_{\mathbb{R}^n} \sum_{i,j} \text{tr}\big(F_{ij}(x)^2\big) \ dx.$$

We now proceed to define further structure in our Yang-Mills theory. Denote

$$d_A \omega := d\omega + [A \wedge \omega], d_A^* \eta = (-1)^{n(k-1)1} * d_A * \eta$$

for a k-form ω. Note that F satisfies the Bianchi Identity, that is

$$d_A F = 0$$

and d_A^* is the adjoint operator to d_A, or

$$\int_M \text{tr}(\langle d_A \eta, \gamma \rangle) \omega = \int_M \text{tr}(\langle \eta, d_A^* \gamma \rangle) \omega.$$

The adjoint operator is also particularly useful since we can defined the differential element of the action as

$$\lim_{t\to 0}\frac{S(A+tB)-S(A)}{t} = -\int_M \mathrm{tr}(B\wedge *d_A^* F_A).$$

In addition, A is a **Yang-Mills Connection** if and only if $d_A^* F_A \equiv 0$, or satisfies the Yang-Mills equation.

Let's now consider a case $G = U(1)$ so that $F_A = dA$. Note that if A is exact, then $dA = d^2 f = 0$, so A is a Yang-Mills connection. The case of $U(1)$ simplifies the Yang-Mills Equation to $d * F = 0$. In coordinates, we would have

$$*F_{ij} = \mathrm{sgn}(\pi)F_{kl}$$

where π is a permutation of $1, 2, 3, 4$. This implies that

$$\sum_{j=1}^{4}\frac{\partial F_{ij}}{\partial x_j} = 0 \;\forall\; i$$

or

$$\sum_{j=2}^{4}\frac{\partial^2 A_j}{\partial x_j^2} = 0$$

of which the second equation yields Laplace's equation, naturally suggesting a relation to E&M theory we will discuss in the next section.

4.0.1 Hints of Gauge Theory

Importantly, in a physical theory, the gauge field is not the physically observable quantity, but rather the curvature form. In fact, different choices of gauge fields give rise to the same curvature form, so we want to actually study $\mathcal{A}/\mathcal{G}$, and choose a representative from each equivalent class. Denote $\mathcal{G} = \{\text{gauge transforms } C^\infty(M, G)\}$ and $\mathcal{A} = \{\text{gauge fields } \Omega^1(M, \mathfrak{g})\}$. Note that $\mathcal{G}$ has a group structure via $(U_1U_2)(x) := U_1(x)U_2(x)$ for $U_1, U_2 \in \mathcal{G}$. In the rest of this section, we denote multiplication of forms as $\eta\gamma := \eta\wedge\gamma$. Thus, $(dU)U^{-1}$ is a $\mathfrak{g}$-valued 1-form. In addition, define a gauge transform by $U \in \mathcal{G}$ as

$$A^U := UAU^{-1} - (dU)U^{-1}$$

with group action $(A^U)^V = A^{VU}$. We therefore define A to be **gauge equivalent** to B if there exists $U \in \mathcal{G}$ such that $B = A^U$ so that $A, B \in \mathcal{A}/\mathcal{G}$. Note that a function f is **gauge invariant** if $f(A^U) = f(A)$ for all $A \in \mathcal{A}, U \in \mathcal{G}$. In particular, the Yang-Mills action is gauge invariant. To see this, note that if $B = A^U$, then $F_B = UF_AU^{-1}$, and writing computations down yields that $S(B) = S(A)$.

In the case of $U(1)$ gauge theory, elements of $\mathcal{G}$ are of the form $e^{i\psi(x)}$. Thus, we have that $(dU)U^{-1} = i\sum_{i=0}^{3}\frac{\partial\psi}{\partial x_i}dx_i = id\psi$ which implies that a gauge transform is just

$$A^\psi = A - id\psi.$$

4.1 U(1) and Electrodynamics

Let us now focus on $U(1)$ gauge theory further, developing the physical theory of electrodynamics along with a rigorous construction of the quantization of the free electromagnetic field in quantum electrodynamics. To recall, we have our gauge field $A : \mathbb{R}^4 \to (i\mathbb{R})^4$ with action (when written in local coordinates)

$$S(A) = \frac{1}{2}\int_{\mathbb{R}}\int_{\mathbb{R}^3}\left(\sum_{j=1}^{3}\left(\frac{\partial A_j}{\partial t} - \frac{\partial A_0}{\partial x_j}\right)^2 - \sum_{1\leq j\leq k\leq 3}\left(\frac{\partial A_k}{\partial x_j} - \frac{\partial A_j}{\partial x_k}\right)^2\right)\, dxdt.$$

Define

$$\mathcal{A}_s = \{U(1) \text{ connections on } \mathbb{R}^4 \text{ such that } A_j \in \mathcal{S}(\mathbb{R}^4, \mathbb{C})\}$$

$$\mathcal{G}_s = \{\text{Schwartzian gauge transforms}\}$$

with $\mathcal{G}_s$ a group action on $\mathcal{A}_s$. In electrodynamics, we choose the Lorenz gague on $\mathcal{A}_s/\mathcal{G}_s$ so that

$$-\frac{\partial A_0}{\partial t} + \sum_{j=1}^{3}\frac{\partial A_j}{\partial x_j} = 0$$

with the set

$$\mathcal{A}_s^L := \{A \in \mathcal{A}_s \text{ such that } A \text{ is Lorenz}\}.$$

Thus, we have that $\Phi(A, \psi) := A^\psi$ an isomorphism of vector spaces $\mathcal{A}_s^L \times \mathcal{G}_s$ and $\mathcal{A}_s$. Therefore, we can write A uniquely as some B^ψ so that

$$\tilde{S}(A) = S(B) + F(\psi)$$

where $F(\psi)$ is the **Feynman gauge fixing term** defined by

$$F(\psi) := \frac{1}{2}\int_{\mathbb{R}}\int_{\mathbb{R}^3}(\Box\psi(t,x))^2\, dxdt$$

with gauge transform

$$\tilde{S}(A) = S(A) - \frac{1}{2}\int_{\mathbb{R}}\int_{\mathbb{R}^3}\left(\frac{\partial A_0}{\partial t} - \sum_{j=1}^{3}\frac{\partial A_j}{\partial x_j}\right)^2\, dxdt.$$

This gives the full action

$$\tilde{S}(A) = -\frac{1}{2}\int_{\mathbb{R}}\int_{\mathbb{R}^3}((\partial_t A_0(t,x))^2 - ||\nabla A_0(t,x)||^2)\, dxdt + \frac{1}{2}\sum_{j=1}^{3}\int_{\mathbb{R}}\int_{\mathbb{R}^3}((\partial_t A_j(t,x))^2 - ||\nabla A_j(t,x)||^2\, dxdt.$$

To construct the free electromagnetic field, we Wick rotate the above action to get

$$\tilde{S}_E(A) = -\frac{1}{2}\int_{\mathbb{R}}\int_{\mathbb{R}^3}((\partial_t A_0(t,x))^2 - ||\nabla A_0(t,x)||^2)\,dxdt + \frac{1}{2}\sum_{j=1}^{3}\int_{\mathbb{R}}\int_{\mathbb{R}^3}((\partial_t A_j(t,x))^2 - ||\nabla A_j(t,x)||^2\,dxdt$$

Which gives A^E a random field on $\mathbb{R}^4$ with probability density proportional to $e^{-\tilde{S}_E(A)}$. In this case, such a random field would be satisfied by $A_0^E = i\phi_0$ and $A_j^E = \phi_j$. Thus, $\{A^E(t,\cdot)\}_{t\in\mathbb{R}}$ is a stationary, reversible, Markov process with state space

$$\mathcal{U} := (i\mathbb{R})^{\mathcal{S}(\mathbb{R}^3,\mathbb{R})} \times \mathbb{R}^{\mathcal{S}(\mathbb{R}^3,\mathbb{R})} \times \mathbb{R}^{\mathcal{S}(\mathbb{R}^3,\mathbb{R})} \times \mathbb{R}^{\mathcal{S}(\mathbb{R}^3,\mathbb{R})}$$

This gives now the Hilbert space structure for our QFT with $\mathcal{H} = L^2(\mu)$ and H, the Hamiltonian, the generator of the semigroup $(P(t))_{t\geq 0}$ on $L^2(\mu)$ given by $P(t)F(x) := \mathbb{E}(F(A^E(t,\cdot))|A^E(0,\cdot) = X)$ and vacuum state $\Omega \equiv 1$ so that $H\Omega = 0$.

4.1.1 Photon Propagator

Define the photon propagator as

$$\Pi = (\Pi_{jk})_{1\leq j,k\leq 3}$$

defined by

$$\Pi_{jk}(x-y) := i\langle\Omega|\mathcal{T}\{A_j(x)A_k(y)\}|\Omega\rangle$$

which written out gives

$$\Pi_{jk}(x-y) = -\frac{i\eta_{jk}}{(2\pi)^3}\int_{\mathbb{R}^3}\frac{e^{-i|t|||p||+ip\cdot x}}{2||p||}\,dp$$

which can be thought of as a functional on Schwartz functions as

$$\Pi_{jk}(t,f,g) := -\frac{i\eta_{jk}}{(2\pi)^3}\int_{\mathbb{R}^3}\frac{e^{-i|t|||p||}\hat{f}(p)\overline{\hat{g}(p)}}{2||p||}\,dp.$$

This propagator relates to the gauge field via

$$\langle\Omega|A_j(s,g)A_k(t,f)|\Omega\rangle = -i\Pi_{jk}(t-s,f,g)$$

In momentum coordinates, we have that

$$\Pi_{jk}(x) = \lim_{\epsilon\downarrow 0}\frac{\eta_{jk}}{(2\pi)^4}\int_{\mathbb{R}^4}\frac{e^{-i(x,p)}}{p^2+i\epsilon}\,dp$$

so that

$$\tilde{\Pi}_{jk}(p,q) := \frac{i}{(2\pi)^4}\langle\tilde{A}_j(p)\tilde{A}_k(q)\rangle = \eta_{jk}\delta^{(4)}(p+q)\lim_{\epsilon\downarrow 0}\frac{1}{q^2+i\epsilon}.$$

4.2 Grassman Variables

Grassman variables are constructed from the set generated by $\psi_1, ..., \psi_n$ quotiented out by the ideal $\psi_i\psi_j + \psi_j\psi_i = 0$. We can also integrate and differentiate Grassman variables using the following rules:

$$\frac{d}{d\psi}\psi = 1, \frac{d}{d\psi}1 = 0.$$

Note that differentiation is noncommutative, since

$$\frac{d}{d\psi_1}\frac{d}{d\psi_2}\psi_2\psi_1 = 1, \frac{d}{d\psi_2}\frac{d}{d\psi_1}\psi_2\psi_1 = \frac{d}{d\psi_2}\frac{d}{d\psi_1}(-\psi_1\psi_2) = -1$$

Integration of variables is also defined in the following way:

$$\int \psi \; d\psi = 1, \int d\psi = 0$$

so in fact we see that for any function of Grassman variables f, we have that

$$\int f \; d\psi := \frac{df}{d\psi}$$

which makes sense since all functions must only contain at most terms of linear order in any Grassman variable.

Theorem 4.1 (Gaussian Grassman Integral). For any complex-valued matrix $M = (m_{ij})$, we have that

$$\int \exp\left(-\sum_{i,j=1}^{n} m_{ij}\overline{\psi_i}\psi_j\right) d\psi d\overline{\psi} = \det M.$$

For Grassman-valued random variables $(X, \overline{X}) = (X_1, ..., X_n, \overline{X_1}, ..., \overline{X_n})$ with mean $\mu = 0$ and covariance matrix M, we have that

$$\mathbb{E}(F(X, \overline{X})) := \frac{\int F(\psi, \overline{\psi}) \exp\left(-\sum_{i,j=1}^{n} m_{ij}\overline{\psi_i}\psi_j\right) \; d\psi \; d\overline{\psi}}{\int \exp\left(-\sum_{i,j=1}^{n} m_{ij}\overline{\psi_i}\psi_j\right) \; d\psi d\overline{\psi}}.$$

4.3 Fadeev-Popov Ghosts

Due to the nature of gauge-fixing, we need a way to concretely capture the gauge freedom in path integrals. Consider a function $f : S \to \mathbb{C}$ so that for all $g \in G$, the Lie group, we have that $f(x^g) = f(x)$ for all $x \in S$ and $\mu(A^g) = \mu(A)$ for all $g \in G, A \subseteq S$ where $A^g := \{x^g : x \in A\}$. Now, supose $T \subseteq S$ has an element in each orbit for all $x \in S$, let $\pi(x)$ be the representative of the orbit in T. Let ν be the pushforward of μ by π, or $\nu(A) := \mu(\pi^{-1}(A))$ for $A \subseteq T$. Then, we have that

$$\int_S f(x) \; d\mu(x) = \int_T f(y) d\nu(y).$$

As an example, let $T = \{(x, 0) : x \geq 0\}$ and $f : \mathbb{R}^2 \to \mathbb{C}$ with G being the set of rotations in $\mathbb{R}^2$. Then, we have that

$$\int_{\mathbb{R}^2} f(x, y)\ dxdy = 2\pi \int_0^\infty f(r, 0)\ dr$$

where $d\nu(r) = 2\pi r d\mu_0(r)$ which intuitively captures radial symmetry in f.

Remark 4.2. There are some assumptions present to ensure the following theorem can be rigorously justified. In particular, we require the action $x \mapsto x^g$ to preserve the Lebesgue measure, and $(g, x) \mapsto x^g$ to be smooth. $\Psi : \mathfrak{g} \times \mathbb{R}^n \to \mathbb{R}^n$ is a map such that

$$\Psi(X, x) := \Phi(x^{e^X})$$

where Φ is surjective and linear. In addition, for all $x \in \mathbb{R}^n$, O_x, the orbit of x under G intersects $\ker \Phi$, and there are distinct $g_1, ..., g_k \in G$ so that for almost all $y \in \ker \Phi$, we have $\{g \in G : y^g \in \ker \Phi\} = \{g_1, ..., g_k\}$. Thus, rewriting $X = \sum_{j=1}^m X_j T_j$ for an orthonormal basis T_j of $\mathfrak{g}$, we let

$$\partial_X \Psi := (\partial_{X_a} \Psi_b).$$

Theorem 4.3 (Fadeev-Popov). For gauge invariant $F \in C(\mathbb{R}^n) \cap L^1(\mathbb{R}^n)$, we have that

$$\int_{\mathbb{R}^n} F(x)\ dx = \frac{\mu(G)}{k\Delta} \int_{\ker \Phi} F(y) |\det(\partial_X \Psi(0, y))|\ dy$$

where μ is the unnormalized Haar measure on G, and $\Delta := |\det(\Phi|_{(\ker \Phi)^\perp})|$. Additionally, for $\gamma_1, ..., \gamma_m, \overline{\gamma_1}, ..., \overline{\gamma_m}$ Grassman variables, we have that

$$\int_{\mathbb{R}^n} F(x)\ dx = \frac{\mu(G)}{k\Delta} \int_{\ker \Phi} \int F(y) \exp\left(i\pi\sigma(y) - \sum_{i,j=1}^m \partial_{X_a} \Psi_j(0, y) \overline{\gamma_j} \gamma_a \right)\ d\gamma d\overline{\gamma} dy$$

where $d\gamma d\overline{\gamma} = d\gamma_1 d\overline{\gamma_1} ... d\gamma_m d\overline{\gamma_m}$ and $\sigma(y) = 1$ if $\det(\partial_X \Psi(0, y)) < 0$ and 0 otherwise.

5 Wightman Axioms

Wightman developed a series of axioms so that one can have a quantum field theory consistent with special relativity and quantum mechanics with no physical inconsistencies. First, let us review the Poincare group, which is the semidirect product

$$\mathcal{P} = \mathbb{R}^{1,3} \rtimes SO^+(1,3) = \{(a, A) : a \in \mathbb{R}^{1,3}, A \in SO^+(1,3)\}$$

with group operation

$$(a, A)(b, B) = (a + Ab, AB)$$

Axiom 5.1 (Wightman Axioms)**.**

1. There exists $\mathcal{H}$ such that for all states $\psi \in \mathcal{H}$ and for unitary representations U of $\mathcal{P}$ in $\mathcal{H}$ so that by the Spectral theorem,

$$U(a, 1) = \int_{\mathbb{R}^{1,3}} e^{i(a,p)} dE(p)$$

where E is a projection-valued measure on $\mathbb{R}^{1,3}$ and the representations satisfy

$$U((a, A)(b, B)) = U(a, A)U(b, B).$$

2. There exists a dense subspace $\mathcal{D}$ of $\mathcal{H}$ and a linear map

$$\phi : \mathcal{S}(\mathbb{R}^4, \mathbb{C}) \to \{\text{unbounded operators on } \mathcal{H}\}$$

such that for all $f \in \mathcal{S}(\mathbb{R}^4, \mathbb{C})$, $\phi(f)$ is defined on $\mathcal{D}$ and $\phi(f)(\mathcal{D}) \subseteq \mathcal{D}$ such that for all $\xi, \eta \in \mathcal{D}$,

$$(\phi(\overline{f})\xi, \eta) = (\xi, \phi(f)\eta).$$

3. For $\xi, \eta \in \mathcal{D}$, the mapping $f \mapsto (\xi, \phi(f)\eta)$ is tempered.
4. For all $(a, A) \in \mathcal{P}$,

$$U(a, A)\phi(f)U(a, A)^{-1} = \phi(V(a, A)f)$$

where

$$V(a, A)f(x) = f(A^{-1}(x - a)).$$

5. (Locality) For all $f, g \in \mathcal{S}(\mathbb{R}^4, \mathbb{C})$ such that $\text{supp}(f) \cap \text{supp}(g) = \emptyset$, then $[\phi(f), \phi(g)] = 0$. with spacelike support separation, so $(x - y)^2 < 0$.
6. (Existence and Uniqueness of Vacuum Sates) There exists $\Omega \in \mathcal{D}$ such that $U(a, A)\Omega = \Omega \; \forall \; (a, A) \in \mathcal{P}$.
7. The closed span of $\phi(f_1)...\phi(f_n)\Omega$ and $f_1, ..., f_n$ is equal to $\mathcal{H}$.

6 Lattige Gauge Theory, d=2

Now that we have gone over the preliminaries for rigorous constructions of quantum fields, it is best to describe a framework of quantum fields on a lattice. Let G be a compact matrix Lie group with $G \subseteq U(n)$ and $\mathfrak{g}$ the corresponding Lie algebra. As in the framework of Yang-Mills, we work over $\mathbb{R}^4$ with gauge field A so that

$$F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j]$$

$$S(A) = -\frac{1}{2g_0^2} \int_{\mathbb{R}^4} \mathrm{Tr} \left(\sum_{0 \le i < j \le 3} \eta_i \eta_j F_{ij}(x)^2 \right) \, dx.$$

Consider a Wick Rotation:

$$dx_0 \mapsto -i dx_0, \partial_0 \mapsto i\partial_0, A_0 \mapsto iA_0$$

so that the Euclidean action is

$$S_E(A) := -\frac{1}{2g_0^2} \int_{\mathbb{R}^4} \mathrm{Tr} \left(\sum_{0 \le i < j \le 3} F_{ij}(x)^2 \right) \, dx$$

which gives a Probability measure on $\mathfrak{g}$-valued 1-forms on $\mathbb{R}^4$ with probability density proportional to $e^{-S_E(A)}$.

Now, let $\Lambda \subseteq \mathbb{Z}^d$ such that for all adjacent vertices $x, y \in \Lambda$, there exists $U(x, y) \in G$ such that $U(x, y) = U(y, x)^{-1}$. This is called a **configuration**, and denote

$$G(\Lambda) = \{\text{configurations}\}$$

Denote a **plaquette** by a square bounded by four edges. Let

$$p(\Lambda) = \{\text{plaquettes}\}$$

With unitary operator on the plaquettes as

$$U_p := U(x_1, x_2)U(x_2, x_3)U(x_3, x_4)U(x_4, x_1)$$

where $x_1, x_2, x_3, x_4 \in \Lambda$ are the 4 vertices in counterclockwise order for the plaquette. This allows us to define the Wilson Action on Λ

$$S_\Lambda(U) = \sum_{p \in p(\Lambda)} \mathrm{Re}(\mathrm{Tr}(I - U_p))$$

This provides a normalized product Haar measure on $G(\Lambda)$, σ_Λ. For $\beta > 0$, we get a probability measure $\mu_{\Lambda,\beta}$ such that

$$d\mu_{\Lambda,\beta}(U) := \frac{1}{Z} e^{-\beta S_\Lambda(U)} \, d\sigma_\Lambda(U)$$

which is called the **Lattice Gauge Theory** with gauge group G, inverse coupling strength β, and free boundary condition.

Theorem 6.1 (Lattice Gauge Theory Relation to Yang-Mills Theory)**.** For $d \geq 2$ and $A = \sum_{j=1}^{d} A_j dx_j$ and $U(x, x+\epsilon e_j) = e^{\epsilon A_j(x)}, U(x+\epsilon e_j, x) = U(x, x+\epsilon e_j)^{-1}$ and plaquette p in $\epsilon\mathbb{Z}^d$, we have that

$$\lim_{\epsilon \to 0} g_0^{-2} \epsilon^{d-4} S(U) = S_E(A)$$

6.1 Gauge Transforms

On the trivial principle bundle $\mathbb{R}^d \times G$, we have gauge transforms $U : \mathbb{R}^d \to G$ such that the induced group action is

$$A^U := UAU^{-1} - (dU)U^{-1}$$

Now, consider some $\Lambda \subseteq \mathbb{Z}^d, G^\Lambda = \{\text{maps: } \Lambda \to G\}$. A gauge transform g acts on U as

$$U^g(x, y) := g(x)U(x, y)g(y)^{-1}$$

Notably, $U^g \in G(\Lambda)$ since

$$\begin{aligned} U^g(y, x) &= g(y)U(y, x)g(x)^{-1} \\ &= g(y)U(x, y)^{-1}g(x)^{-1}(g(x)U(x, y)g(y)^{-1})^{-1} \\ &= U^g(x, y)^{-1} \end{aligned}$$

Theorem 6.2. Let A be a smooth, compactly supported $\mathfrak{g}$-valued 1-form and $g \in C^\infty(\mathbb{R}^d, G)$ equal to I outside of a compact set. For $x \in \epsilon\mathbb{Z}^d$, we have for $B_j(x) := \epsilon^{-1} \log U^g(x, x+\epsilon e_j)$, then

$$\lim_{\epsilon \to 0} \sup_{x \in \epsilon\mathbb{Z}^d, 1 \leq j \leq d} ||A_j^g(x) - B_j(x)|| = 0.$$

We now wish to compute vacuum expectation values in quantum Yang-Mills theory for observables $f : G(\Lambda) \to \mathbb{C}$. For a gauge invariant f, we have that

$$\forall\, U \in G(\Lambda), g \in G^\Lambda \implies f(U^g) = f(U).$$

A **Wilson Loop** is a sequence of points $x_0, x_1, ..., x_k \in \mathbb{Z}^d$ such that for all $0 \leq j \leq k-1$, we have that x_{j+1} and x_j are neighbors along with x_0 and x_k. Define

$$W_l(U) := \text{Tr}(U(x_0, x_1)U(x_1, x_2)...U(x_{k-1}, x_k)U(x_k, x_0)).$$

Note that this is a gauge invariant observable since $W_l(U^g) = W_l(U)$. For p plaquette in Λ,

the mapping $U \mapsto \mathrm{Tr}(U_p)$ itself is a Wilson loop observable with p a loop. Putting this in the Yang-Mills action, we see that it itself is gauge invariant. This further implies that the measure $\mu_{\Lambda,\beta}$ is invariant under gauge transforms since for all $f : G(\Lambda) \to \mathbb{C}$ and $g \in G^\Lambda$,

$$\int_{G(\Lambda)} f(U)\ d\mu_{\Lambda,\beta}(U) = \int_{G(\Lambda)} f(U^g)\ d\mu_{\Lambda,\beta}(U)$$

as σ_Λ is invariant under gauge transformations.

For gauge fixing, define $G(\Lambda, \Gamma) = \{\text{subset of configurations } U \in G(\Lambda)\}$ such that $U(x, y) = I\ \forall\ (x, y) \in \Gamma$. Define the lattice gauge theory here as

$$d\mu_{\Lambda,\Gamma,\beta}(U) := \frac{1}{Z_\Gamma} e^{-\beta S_\Lambda(U)} d\sigma_{\Lambda,\Gamma}(U)$$

where

$$d\sigma_{\Lambda,\Gamma(U)} = \prod_{(x,y)\in E(\Lambda)\setminus\Gamma} d\sigma(U(x, y)).$$

Thus, for all gauge invariant observable $f : G(\Lambda) \to \mathbb{C}$ such that $\langle |f| \rangle < \infty$, then we have that

$$\langle f \rangle = \int_{G(\Lambda,\Gamma)} f(U) d\mu_{\Lambda,\Gamma,\beta}(U).$$

6.2 Mass Gaps and Quark Confinement

Consider a sequence $\Lambda_N \subseteq \mathbb{Z}^d$ such that $\Lambda_N \uparrow \mathbb{Z}^d$ as $N \to \infty$. A **mass gap** exists when for f a gauge invariant observable supported on E of edges, then it does not depend on $U(x, y)$ for all $(x, y) \notin E$. Such an example of this is the Wilson loop observable. In particular, for $S_1, S_2 \subseteq \Lambda_N$ and f_1, f_2 bounded gauge invariant observables supported on S_1, S_2 respectively, we have that

$$|\langle f_1 f_2 \rangle - \langle f_1 \rangle \langle f_2 \rangle| \leq C_1 e^{-C_2 \mathrm{dist}(S_1,S_2)}$$

Quark Confinement exists for a function $V : (0, \infty) \to \mathbb{R}$ such that $\lim_{x\to\infty} V(x) = \infty$ and for all rectangular loops l in $\mathbb{Z}^d$ with side lengths R and T such that $R \leq T$, then

$$|\langle W_l \rangle| \leq C e^{-V(R)T}$$

for $V(R)$ proportional to R, we get Wilson's area law which decays as fast as $\exp(-c \cdot \mathrm{area}(l))$. An example of a deconfining theory is the perimeter law, where

$$\langle W_l \rangle \sim C_1 e^{-C_2(R+T)}$$

Theorem 6.3. For $d = 2$, there always exists a mass gap in all Lattice Gauge Theories.

Part II

Algebraic Quantum Field Theory

7 $\mathbf{C}^*$*Algebras*

Sections 12-14 follow [FR19], see more for further discussion. C^* algebras are a naturally setting to consider quantum field theory purely algebraically, that is without considering the underlying Hilbert space structure. The relevance of this is particularly important due to the arbitrariness of the Hilbert space depending on the quantum system (e.g. standard quantum mechanics in a finite-dimensional space as opposed to quantum fields with infinite-dimensional Hilbert spaces). The GNS construction will show that the Hilbert space naturally follows from the given C^* algebra we impose on our system, which we will describe in this section. This allows us to consider the algebraic properties of our QFT, and describe axioms that are formulated for a QFT purely algebraically, the Haag-Kastler axioms. First, let us cover some fundamentals first.

Definition 7.1. A ***-algebra** $\mathcal{A}$ is an algebra over $\mathbb{C}$ such that given a map $* : \mathcal{A} \to \mathcal{A}$ called the **involution map**, we have the following properties:

$$\forall\ A, B \in \mathcal{A}, (A+B)^* = A^* + B^*, (AB)^* = B^*A^*$$

$$\forall\ \lambda \in \mathbb{C}, A \in \mathcal{A}, (\lambda A)^* = \overline{\lambda} A^*$$

$$\forall\ A \in \mathcal{A}, (A^*)^* = A$$

In addition, homomorphisms between $*$-algebras preserve the property of involution and preserve units if the $*$-algebra is **unital**.

A canonical example of a $*$-algebra is $\mathcal{B}(\mathcal{H})$, the space of bounded operators on $\mathcal{H}$ a Hilbert space, and involutions are simply applying the adjoint operator.

Definition 7.2. A normed algebra $\mathcal{A}$ is an algebra equipped with a norm $||\cdot||$ such that

$$||AB|| \leq ||A||||B||$$

If $\mathcal{A}$ is a $*$-algebra where $||A^*|| = ||A||$, and $\mathcal{A}$ is complete under the topology induced by this norm, we call $\mathcal{A}$ a B^*-algebra. Normed algebras where only the last condition is defined is called a Banach Algebra. Lastly, if all of the above properties exist for $\mathcal{A}$ such that

$$||A^*A|| = ||A||||A^*|| = ||A||^2$$

then $\mathcal{A}$ is a $\mathbf{C}^* - algebra$

Indeed, under the operator norm, $\mathcal{B}(\mathcal{H})$ is a C^*-algebra. Given a state ω, we also assign $\omega(A)$ an expectation value. Examples come from unit vectors $\psi \in \mathcal{H}$ with $\mathcal{A} = \mathcal{B}(\mathcal{H})$ such that

$$\omega_\psi(A) = \langle \psi | A\psi \rangle$$

or via a density matrix so that

$$\omega_\rho(A) = \text{tr}(\rho A)$$

which is shown by proving $\omega(\mathbb{1}) = 1$ and $\omega(A^*A) \geq 0$.

Definition 7.3. A **representation** of a unital $*$-algebra $\mathcal{A}$ consists of a triple $(\mathcal{H}, \mathcal{D}, \pi)$ such that $\mathcal{H}$ is a Hilbert space, $\mathcal{D}$ is a dense subspace of $\mathcal{H}$, and $\pi : \mathcal{A} \to Op(\mathcal{H})$ where $Op(A)$ denotes operators on A such that $\pi(A)$ has domain $\mathcal{D}$ and range in $\mathcal{D}$, $\pi(\mathbb{1}) = \mathbb{1}|_{\mathcal{D}}$, π is linear, and $\pi(A^*) = \pi(A)^*|_{\mathcal{D}}$.

Unitary equivalence between two representations $(\mathcal{H}_1, \mathcal{D}_1, \pi_1)$ and $(\mathcal{H}_2, \mathcal{D}_2, \pi_2)$ of a $*$-algebra $\mathcal{A}$ means that there exists some unitary map $U : \mathcal{H}_1 \to \mathcal{H}_2$ that is an isomorphism between $\mathcal{D}_1$ and $\mathcal{D}_2$ such that

$$U\pi_1(A)U^{-1} = \pi_2(A)$$

We are now ready to see the analogies between algebraic states and how they function in Hilbert spaces.

Theorem 7.4 (Gelfand-Naimark-Segal Construction Theorem). Given a state ω on a C^*-algebra $\mathcal{A}$, one can construct a triplet $(\mathcal{H}_\omega, \Omega_\omega, \pi_\omega)$ of $\mathcal{A}$ where $\mathcal{H}_\omega$ is a Hilbert space, $\Omega_\omega \in \mathcal{H}_\omega$ is a unit vector $\Omega_\omega \in \mathcal{D}_\omega$, and π_ω is a representation of $\mathcal{A}$. Additionally, we have that $\pi_\omega(\mathcal{A})\Omega_\omega$ and $\pi_\omega(A)$ extends to a bounded operator on $\mathcal{H}_\omega$. ω also satisfies the expectation value behavior we want, that is

$$\omega(A) = \langle \Omega_\omega | \pi_\omega(A) \Omega_\omega \rangle$$

for any $A \in \mathcal{A}$. In addition, this triplet is unique up to unitary equivalence.

Proof. Define $\mathcal{N}_\omega \subset \mathcal{A}$ by

$$\mathcal{N}_\omega = \{A \in \mathcal{A} : \omega(A^*A) = 0\}$$

We show this is a left ideal using the Cauchy-Schwarz identity. We have that for $A \in \mathcal{I}_\omega$ and $B \in \mathcal{A}$, then

$$\begin{aligned} |\omega((BA)^*(BA))|^2 &= |\omega((B^*BA)^*A)|^2 \\ &\leq \omega((B^*BA)^*(B^*BA))\omega(A^*A) \\ &= 0 \end{aligned}$$

which implies that $BA \in \mathcal{I}_\omega$. In addition, $\mathcal{I}_\omega^*$ is a right ideal, since for $A^* \in \mathcal{I}_\omega$

$$\begin{aligned}|\omega((A^*B)(A^*B)^*)|^2 &= |\omega((BB^*A)^*A)|^2 \\ &\leq \omega((BB^*A)^*(BB^*A))\omega(A^*A) \\ &= \omega((BB^*A)^*(BB^*A))\overline{\omega(AA^*)} \\ &= 0\end{aligned}$$

Now, define the quotient vector space $\mathcal{D}_\omega := \mathcal{A}/\mathcal{I}_\omega$. with a well-defined inner product and norm

$$\langle [A], [B] \rangle_\omega = \omega(A^*B), \quad ||[A]||_\omega^2 = \omega(A^*A)$$

since for all $A \in \mathcal{N}_\omega$, we have that

$$|\omega(A^*B)|^2 \leq \omega(A^*A)\omega(B^*B) = 0.$$

Thus, by linearity of the inner product, all elements in the same equivalence class have the same inner product when taken against any $B \in \mathcal{A}$. Under this inner product, we get a completion of $\mathcal{D}_\omega$ as $\mathcal{H}_\omega$.

The representation π_ω is defined such that for all $A, B \in \mathcal{A}$, we have

$$\pi_\omega(B)[A] = [BA], \quad \pi_\omega(\mathbb{1}) = I.$$

It is easy to show that π is linear. We also have that for all $A, B, C \in \mathcal{A}$

$$\begin{aligned}\langle [C], \pi_\omega(A)[B] \rangle_\omega &= \langle [C], [AB] \rangle_\omega \\ &= \omega(C^*AB) \\ &= \omega((A^*C)^*B) \\ &= \langle [A^*C], [B] \rangle_\omega \\ &= \langle \pi_\omega(A^*)[C], [B] \rangle_\omega\end{aligned}$$

which implies that $\pi_\omega(A)^* = \pi_\omega(A^*)$. We also need to show that $\pi_\omega(A)$ extends to a bounded linear operator for all $A \in \mathcal{A}$. To do this, we look at the spectrum of A^*A. Note that both A^*A and B^*A^*AB are self adjoint, and since all values in $\sigma(A^*A)$ lie in the spectrum between 0 and $||A||^2$, then all values in the spectrum of $||A||^2 I - A^*A$ are nonnegative. Since this state is itself self-adjoint, then for all $B \in \mathcal{A}$, we have that $B^*(||A||^2 - A^*A)B$ has nonnegative spectrum. Along with Cauchy-Schwarz, we see that

$$\begin{aligned}||\pi_\omega(A)[B]||_\omega^2 &= |||[AB]_\omega||^2 \\ &= \omega(B^*A^*AB) \\ &\leq \omega(B^*B||A||^2) \\ &= ||A||^2\omega(B^*B) \\ &= ||A||^2||[B]||_\omega^2.\end{aligned}$$

We also now can define $\Omega_\omega = [\mathbb{1}]_\omega$ since $||\Omega_\omega||^2 = \omega(\mathbb{1}) = 1$. Additionally, we now have each state in the Hilbert space defined uniquely by the vacuum state and the representation since $\pi_\omega(A)\Omega_\omega := [A]_\omega$. Note that this also satisfies

$$\omega(A) = \langle \Omega_\omega, [A]_\omega \rangle_\omega = \langle \Omega_\omega, \pi_\omega(A)\Omega_\omega \rangle_\omega.$$

Lastly, we want to show uniqueness. Consider an alternative triplet (M, ρ, Ψ) with $\omega(A) = \langle \Psi, \rho(A)\Psi \rangle$ for all $A \in \mathcal{A}$. Define a map on states so that for all $A, B \in \mathcal{A}$

$$U(\pi_\omega(A)\Omega_\omega) = \rho(A)\Psi$$

which implies that

$$\begin{aligned} \langle \rho(A)\Psi, \rho(B)\Psi \rangle &= \langle \Psi, \rho(A^*B)\Psi \rangle \\ &= \omega(A^*B) \\ &= \langle [A], [B] \rangle_\omega \\ &= \langle \pi_\omega(A)\Omega_\omega, \pi_\omega(B)\Omega_\omega \rangle_\omega \end{aligned}$$

which implies that

$$\begin{aligned} \langle \rho(A)\Psi, \rho(B)\Psi \rangle &= \langle U(\pi_\omega(A)\Omega_\omega), U(\pi_\omega(B)\Omega_\omega) \rangle \\ &= \langle \pi_\omega(A)\Omega_\omega, U^*U(\pi_\omega(B)\Omega_\omega) \rangle \\ &= \langle \pi_\omega(A)\Omega_\omega, \pi_\omega(B)\Omega_\omega \rangle_\omega \end{aligned}$$

which shows that U is unitary, and in fact extends to a unitary operator for the entire Hilbert space $\mathcal{H}_\omega$ as $U\Omega_\omega = \Psi$, so $\pi_\omega(A)\Omega_\omega = \rho(A)\Psi = \rho(A)U\Omega_\omega$, so we have that $U\pi_\omega(A)U^{-1} = \rho(A)$. □

8 Haag-Kastler Axioms

We discuss here the minimal axioms needed for the AQFT framework. Denote $\mathbb{M}$ as Minkowski spacetime.

Axiom 8.1.

1. Local Algebras - For a unital $*$-algebra $\mathcal{A}(\mathbb{M})$ and any open causally convex bounded region $\mathcal{O} \subset \mathbb{M}$, there exists a subalgebra $\mathcal{A}(\mathcal{O})$ containing the unit of $\mathcal{A}(\mathbb{M})$, so that $\{\mathcal{A}(\mathcal{O})\}$ generate $\mathcal{A}(\mathbb{M})$.

2. Isotony - For $\mathcal{O}_1 \subset \mathcal{O}_2$,

$$\mathcal{A}(\mathcal{O}_1) \subset \mathcal{A}(\mathcal{O}_2).$$

3. Causality - Given causally disjoint regions $\mathcal{O}_1, \mathcal{O}_2$, we have that for all $X \in \mathcal{A}(\mathcal{O}_1)$ and

$Y \in \mathcal{A}(\mathcal{O}_2)$

$$[X, Y] = 0.$$

4. Poincare Covariance - Given a transformation ρ within the identity connected component of the Poincare group, we have an automorphism $\alpha(\rho)$ acting on $\mathcal{A}(\mathbb{M})$ such that

$$\alpha(\rho) : \mathcal{A}(\mathcal{O}) \to \mathcal{A}(\rho\mathcal{O}).$$

5. Existence of Dynamics - For $\mathcal{O}_1$ a Cauchy surface such that $\mathcal{O}_1 \subset \mathcal{O}_2$, then we have that

$$\mathcal{A}(\mathcal{O}_1) = \mathcal{A}(\mathcal{O}_2).$$

9 Example 1 - Free Scalar Field

Consider the free scalar field satisfying the Klein Gordon equation

$$P\Phi = (\Box + m^2)\Phi = 0$$

Define the space

$$\varepsilon = C_c^\infty(\mathbb{M}, \mathbb{R})/PC_c^\infty(\mathbb{M}, \mathbb{R})$$

of which the advanced and retarded green propagators E^- and E^+ act on respectively. Recall that

$$E^\pm Pf = f, PE^\pm f = f.$$

So we define an operator

$$E = E^- - E^+$$

so that for any $f \in C_c^\infty(\mathbb{M}, \mathbb{R})$, we have that

$$P(Ef) = PE^- f - PE^+ f = f - f = 0.$$

Therefore, we can create a symplectic form

$$\sigma(f, g) = \int_{\mathbb{M}} f(x)(Eg)(x)dx$$

where $f, g \in \varepsilon$. Thus, (ε, σ) creates a symplectic space for the algebra of observables to act on.

9.1 Axiom 1: Existence of Local Algebras

Refer to [Dap15] for further material and discussion. Given $f, g \in \varepsilon$, we associate **Weyl operators** to each satisfying:

$$W(f)^* = W(-f), \quad W(0) = 1, \quad W(f)W(g) = e^{-\frac{i}{2}\sigma(f,g)}W(f+g)$$

Remark 9.1. Technically speaking, because we have not defined a Hilbert space for the operators to work over, we can consider these operators as just a vector space

$$\mathbb{C}[\{W(f)\}_{f\in\varepsilon}]/\sim$$

where equivalence comes from the third equality relating $W(f)W(g)$ to $W(f+g)$.

Remark 9.2. We can think of these operator elements as unitary operators of the form

$$W(f) = e^{i\Phi(f)}$$

where $\Phi(f)$ is a **smeared field** – given a function $f \in C_c^\infty(\mathbb{R})$ and Φ a quantum field, then

$$\Phi(f) = \int \Phi(x)f(x)\, dx.$$

This makes sense because

$$\begin{aligned} ||W(f)||^2 &= ||W(f)^*W(f)|| \\ &= ||W(-f)W(f)|| \\ &= ||e^{i\sigma(-f,f)}W(0)|| \\ &= ||1|| \\ &= 1. \end{aligned}$$

We wish to show that

$$[\Phi(f), \Phi(g)] = i\sigma(f,g)\mathbb{1}$$

Proof. Note that

$$e^{-\frac{i}{2}\sigma(f,g)}W(f+g) = W(f)W(g) = e^{i\Phi(f)}e^{i\Phi(g)} = e^{i\Phi(f)+i\Phi(g)+i[\Phi(f),\Phi(g)]}$$

by Baker-Campbell-Hausdorff, but we also have that

$$e^{-\frac{i}{2}\sigma(f,g)}W(f+g) = e^{-\frac{i}{2}\sigma(f,g)}e^{i\Phi(f+g)} = e^{-\frac{i}{2}\sigma(f,g)+\Phi(f+g)}$$

where, we use that $-\frac{i}{2}\sigma(f,g)$ is technically a function times the identity operator, and $\Phi(f+g) = \Phi(f) + \Phi(g)$ by linearity of the smeared field operator. Thus, we have by equality the desired result. □

To construct the algebra, we define

$$\mathcal{A}(\mathcal{O}) = C^*(\{W(f) : \text{supp}(f) \subseteq \mathcal{O}\})$$

where $C^*(\cdot)$ denotes the C^*-completion of the set.

9.2 Axioms 2: Isotony

Isotony is immediate because of the definition. Given $\mathcal{O}_1 \subset \mathcal{O}_2$, we have that

$$\{W(f) : \text{supp}(f) \subseteq \mathcal{O}_1\} \subset \{W(f) : \text{supp}(f) \subseteq \mathcal{O}_2\}$$

and so the above still hold for C^*-completions as well.

9.3 Axiom 3: Causality

Consider two spacelike-separated regions $\mathcal{O}_1$ and $\mathcal{O}_2$. Take functions f and g with supports in $\mathcal{O}_1$ and $\mathcal{O}_2$ respectively. Note that in this case, we have that

$$\sigma(f, g) = \int f(x)(Eg)(x)\, dx = 0$$

since $\text{supp}(Eg) \subseteq (J^+(\text{supp}(g)) \cup J^-(\text{supp}(g))) \subseteq J^+(\mathcal{O}_2) \cup J^-(\mathcal{O}_2)$ which is disjoint to $\text{supp}(f) \subseteq \mathcal{O}_1$. Thus, we have that $W(f)W(g) = W(f+g) = W(g+f) = W(g)W(f)$, so all generators of $\mathcal{A}(\mathcal{O}_1)$ commute with those in $\mathcal{A}(\mathcal{O}_2)$ as desired.

9.4 Axiom 4: Poincare Covariance

For a given $x \in \mathbb{M}$, we consider the group action $\rho := (\Lambda, a) \in \mathcal{P}$ in coordinates as

$$\rho \cdot x := \Lambda x + a.$$

We define the operator

$$(T_\rho f)(x) = f(\rho^{-1} x).$$

For $f \in C_c^\infty(\mathbb{M}, \mathbb{R})$, we wish to show the following.

Lemma 9.3. Given $\rho \in \mathcal{P}$, we have that:

$$\text{supp}(T_\rho f) = \rho \cdot \text{supp}(f), \quad T_\rho P = P T_\rho, \quad T_\rho E = E T_\rho.$$

Proof. We unpack definitions. Let $x \in \text{supp}(T_\rho(f))$. We have that

$$\rho^{-1} \cdot x \in \text{supp} f \iff x \in \rho \cdot \text{supp} f.$$

□

We also want to show that

$$T_\rho P = P T_\rho.$$

Proof. This follows from direct computation. It is clear that $T_\rho m^2 = m^2 T_\rho$. We work in coordinates. We have that

$$\begin{aligned}\partial_\mu \partial_\nu (T_\rho f)(x) &= \partial_\mu \partial_\nu (f(\rho^{-1}x)) \\ &= \partial_\mu \partial_\nu f(\Lambda^{-1}(x-a)) \\ &= (\Lambda_\mu^\sigma)^{-1}(\Lambda_\nu^\beta)^{-1}(\partial_\sigma \partial_\beta f)(\rho^{-1}x) \\ &= (\Lambda_\mu^\sigma)^{-1}(\Lambda_\nu^\beta)^{-1} T_\rho(\partial_\sigma \partial_\beta f)(x)\end{aligned}$$

which implies that

$$\begin{aligned}\Box(T_\rho f)(x) &= \eta^{\mu\nu}\partial_\mu \partial_\nu (T_\rho f)(x) \\ &= \eta^{\mu\nu}(\Lambda_\mu^\sigma)^{-1}(\Lambda_\nu^\beta)^{-1} T_\rho(\partial_\sigma \partial_\beta f)(x) \\ &= \eta^{\sigma\beta} T_\rho(\partial_\sigma \partial_\beta f)(x) \\ &= T_\rho(\Box f)(x)\end{aligned}$$

which follows by definitions of the Lorentz metric. □

Lastly, we want to show that

$$T_\rho E = E T_\rho.$$

Proof. This follows by uniqueness of the propagators. Let J^+ denote the causal future, and J^- denote the causal past. We first have that

$$\text{supp}(E^+ f) \subseteq J^+(\text{supp} f)$$

$$\text{supp}(E^- f) \subseteq J^-(\text{supp} f)$$

following definitions of the propagators. We define

$$\tilde{E}^\pm := T_\rho E^\pm T_{\rho^{-1}}.$$

Using that $PT_\rho = T_\rho P$, we see that

$$P\tilde{E}^\pm = PT_\rho E^\pm T_{\rho^{-1}} = T_\rho P E^\pm T_{\rho^{-1}} = T_\rho T_{\rho^{-1}} = \text{Id}.$$

A similar computation yields $\tilde{E}^\pm P = \text{Id}$. Additionally, we have that

$$\begin{aligned}\text{supp}(\tilde{E}^\pm f) &= \rho \cdot \text{supp}(E^\pm T_{\rho^{-1}} f) \\ &\subseteq \rho \cdot J^\pm \text{supp}(T_{\rho^{-1}} f) \\ &= J^\pm \text{supp}(T_{\rho^{-1}} f).\end{aligned}$$

Therefore, by uniqueness of the propagators from these properties, we have that

$$\tilde{E}^{\pm} = E$$

which gives us that

$$T_\rho E = E T_\rho.$$

□

Given these properties with T_ρ, we define a map $\tilde{T}_\rho : \varepsilon \to \varepsilon$ where

$$\tilde{T}_\rho[f] = [T_\rho f]$$

where we use $[\cdot]$ to denote the equivalence class. Well-definedness follows from the following computation:

$$\begin{aligned}
[T_\rho(f + Ph)] &= [T_\rho f + T_\rho P h] \\
&= [T_\rho f + P T_\rho h] \\
&= [T_\rho f].
\end{aligned}$$

Lastly, we want to compute

$$\begin{aligned}
\sigma(T_\rho f, T_\rho g) &= \int (T_\rho f)(x)(E T_\rho g)(x)\, dx \\
&= \int (T_\rho f)(x)(T_\rho E g)(x)\, dx \\
&= \int f(\rho^{-1}x)(Eg)(\rho^{-1}x)\, dx \\
&= \int f(y)(Eg)(y)\, dy \\
&= \sigma(f, g)
\end{aligned}$$

which follows by u-substitution since Lorentz transformations have determinant ± 1. Thus, we can define the automorphism

$$\alpha(\rho)W(f) = W(\tilde{T}_\rho f)$$

to verify that it is an automorphism, we have that

$$\begin{aligned}
\alpha(\rho)(W(f)W(g)) = \alpha(\rho)(e^{-\frac{i}{2}\sigma(f,g)}W(f+g)) = e^{-\frac{i}{2}\sigma(f,g)}\alpha(\rho)W(f+g) = e^{-\frac{i}{2}\sigma(f,g)}W(T_\rho f + T_\rho g) \\
&= e^{-\frac{i}{2}\sigma(T_\rho f, T_\rho g)}W(T_\rho f + T_\rho g) \\
&= W(T_\rho f)W(T_\rho g) \\
&= (\alpha(\rho)W(f))(\alpha(\rho)W(g)).
\end{aligned}$$

Lastly, given generators of $\mathcal{A}(\mathcal{O})$ of the form $W(f)$, we have that $\alpha(\rho)W(f) = W(T_\rho f)$ satisfies

$\text{supp}(T_\rho f) = \rho \cdot \text{supp} f \subseteq \rho\mathcal{O}$, indeed showing that the map satisfies

$$\alpha(\rho) : \mathcal{A}(\mathcal{O}) \to \mathcal{A}(\rho\mathcal{O}).$$

9.5 Axiom 5: Cauchy Surfaces

See [CF08] for further generalizations and disucssions. Let $\mathcal{O}_1$ be a Cauchy surface contained in an open set $\mathcal{O}_2$, and Σ_+ and Σ_- be Cauchy surfaces in different time slices such that $\Sigma_- \subset J^-(\Sigma_+)$ and similarly $\Sigma_+ \subset J^+(\Sigma_-)$ so that

$$J^+(\Sigma_-) \cap J^-(\Sigma_+) \subset \mathcal{O}_1.$$

Consider a smooth function $\chi_\pm \in C^\infty(\mathbb{M})$ where

$$\begin{aligned}
&\chi_-(x) = 0 \ \forall \ x \in J^-(\Sigma_-) \\
&\chi_-(x) = 1 \ \forall \ x \in J^+(\Sigma_+) \\
&\chi_+ := 1 - \chi_-
\end{aligned}$$

Let $h = \chi_+ E^+ f + \chi_- E^- f$, and consider

$$g := f - Ph$$

We verify that h is compactly supported. Notice that the first term vanishes for $x \in J^+(\Sigma_+)$ and similarly the second term vanishes for $x \in J^-(\Sigma_-)$. Additionally, in any causal past for arbitrary $f \in C_c^\infty(\mathcal{O}_2)$, we have that $E^+ f$ vanishes, and for any causal future of f, $E^- f$ vanishes. In other words,

$$\text{supp}(h) \subseteq (J^-(\Sigma_-) \cap J^+(\text{supp}(f))) \cup (J^+(\Sigma_+) \cap J^-(\text{supp}(f))) \subseteq \mathcal{O}_2$$

which is compact. Thus, $h \in C_c^\infty(\mathcal{O}_2)$. It now suffices to show that

$$\text{supp}(g) \subseteq \mathcal{O}_1$$

To do this, we simplify the following (where $[A, B] = AB - BA$, the commutator bracket):

$$\begin{aligned}
Ph &= \chi_+ PE^+ f + \chi_- PE^- f + [P, \chi_+]E^+ f + [P, \chi_-]E^- f \\
&= \chi_+ f + \chi_- f + [P, \chi_+]E^+ f + [P, \chi_-]E^- f \\
&= f + [P, \chi_+]E^+ f + [P, \chi_-]E^- f \\
&= f + [P, \chi_+]E^+ f - [P, \chi_+]E^- f \\
&= f - [P, \chi_+]Ef
\end{aligned}$$

so we have that

$$g = [P, \chi_+]Ef$$

but note that χ_+ is locally constant outside N, so we have that

$$\text{supp}g \subseteq \text{supp}(\partial_\mu \chi_+) \cap \text{supp}(Ef) \subseteq \mathcal{O}_1$$

as desired. Thus, we have generated a function g in the same equivalence class of f with support only in N, so taking the generators of the form $W(g)$, we see that

$$\mathcal{A}(M) \subseteq \mathcal{A}(N)$$

and isotony yields the other direction, showing that this axiom also holds!

10 Example 2 - Dirac Spinor Field

Recall the equation of motion for spinor fields

$$(i\partial\!\!\!/ - m)\psi(x) = 0$$

where $\psi(x)$ take the form

$$\psi(x) = \begin{bmatrix} \psi_L \\ \psi_R \end{bmatrix}$$

and $\partial\!\!\!/ = \gamma^\mu \partial_\mu$. In general, the space of spinor fields acts on what is known as a spinor bundle S, but in Minkowski space, we have that each fiber S_x for $x \in \mathbb{M}$ is isomorphic to $\mathbb{C}^4$ (2 components - one being left-handed and the other being right-handed), so for our case, it suffices to consider them as operators acting on $C^\infty(\mathbb{M}, \mathbb{C}^4)$. However, we wish to use smeared fields to create well-defined operators to act on quantum states in a provided Hilbert space via the GNS construction, that is

$$\psi(f) = \int_{\mathbb{M}} \psi(x) f(x)\, dx$$

where $f \in C^\infty(\mathbb{M}, S^*)$. Now, define

$$D = i\partial\!\!\!/ - m, D^\dagger = -i\partial\!\!\!/ - m$$

so that

$$DD^\dagger = -(\Box + m^2),$$

so much of the theory built from the Klein-Gordon field can be used in the Dirac spinor field case. In particular, we can construct new advanced and retarded propagators using the ones from before. We define

$$S^- = D^\dagger E^-, \quad S^+ = D^\dagger E^+, \quad S = S^+ - S^-$$

so that for all $f \in C^\infty(\mathbb{M}, \mathbb{C}^4)$, we have that

$$S^\pm D f = D S^\pm f = f, \quad \text{supp}(S^\pm f) \subseteq J^\pm(\text{supp}(f)).$$

We also have cospinor fields, dual to the spinor fields $\overline{\psi}$ as operators on $C^\infty(\mathbb{M}, S^*)$. Thus, we

define the smeared field operator

$$\overline{\psi}(g) = \int_{\mathbb{M}} \overline{\psi}(x) g(x)\ dx$$

for $g \in C^\infty(\mathbb{M}, S)$ so that $\psi(x)f(x)$ and $\overline{\psi}(x)g(x)$ can actually be though of as operator-valued. In coordinates we define

$$\psi(x)f(x) = f_\alpha(x)\psi^\alpha(x), \quad \overline{\psi}(x)g(x) = \overline{\psi}_\alpha(x)g^\alpha(x).$$

Lastly, since the equation of motion for the Dirac field requires that $D\psi = D^\dagger\overline{\psi} = 0$, we integrate by parts with the test functions above to impose

$$\psi(D^\dagger f) = \overline{\psi}(Dg) = 0.$$

10.1 Axiom 1: Existence of Local Algebras

To construct our algebra of observables, we define the canonical anticommutation relations

$$\{\psi(f_1), \psi(f_2)\} = 0, \quad \{\overline{\psi}(g_1), \overline{\psi}(g_2)\} = 0, \quad \{\psi(f), \overline{\psi}(g)\} = -i\langle f, Sh\rangle \mathbb{1}$$

Where

$$\langle f, g\rangle = \int_{\mathbb{M}} f(x)(g(x))\ dx$$

Thus, we can now define our algebra as

$$\mathcal{F}(\mathcal{O}) = C^*(\{\psi(f), \overline{\psi}(g) : \text{supp}(f), \text{supp}(g) \subseteq \mathcal{O}\})$$

Remark 10.1. Note that the algebra is actually the free algebra generated by $\{\psi(f), \overline{\psi}(f)\}$ quotiented by the canonical anticommutation relations. The involution operator satisfies

$$(\psi(f))^* = \overline{\psi}(f^\dagger).$$

Thus, the C^* completion follows from the induced unique C^*-norm.

However, this is not an algebra for our observables! Because ψ has a $U(1)$ gauge symmetry, we actually want to consider

$$\mathcal{A}(\mathcal{O}) = \mathcal{F}(\mathcal{O})^{U(1)} = \{\psi(f)\overline{\psi}(g) : \text{supp}(f), \text{supp}(g) \subseteq \mathcal{O}\}$$

as this will satisfy the desired locality constraint.

Remark 10.2. If we wanted to consider the full algebra, $\mathcal{F}(\mathcal{O})$ satisfies graded locality, but not full causality as perscribed in the axiomatic framework of which we will show.

10.2 Axiom 2: Isotony

Isotony, similar to before, is immediate by definition. We have that for $\mathcal{O}_1 \subset \mathcal{O}_2$

$$\{\psi(f), \overline{\psi}(g) : \text{supp}(f), \text{supp}(g) \subseteq \mathcal{O}_1\} \subset \{\psi(f), \overline{\psi}(g) : \text{supp}(f), \text{supp}(g) \subseteq \mathcal{O}_2\}$$

which implies the C^* completion and the algebra of observables also yields the desired inclusion relation.

10.3 Axiom 3: Causality

First, note that given $\mathcal{O}_1$ causally disjoint from $\mathcal{O}_2$, we have that for all $f, g \in C_c^\infty(\mathbb{M}, \mathbb{C}^4)$ such that $\text{supp}(f) \subseteq \mathcal{O}_1, \text{supp}(g) \subseteq \mathcal{O}_2$, then

$$J(\text{supp}(f)) \cap J(\text{supp}(g)) = \emptyset$$

where $J(A) := J^+(A) \cup J^-(A)$. Thus, because $\text{supp}(Sg) \subseteq J(\text{supp}(g))$, then $\langle f, Sg \rangle = 0$ by definition, so all observables anti-commute. Now, all generators of observables have even parity following the anticommutation relations. Thus, we have that

$$[A_1, A_2] = 0$$

for all $A_1, A_2 \in \mathcal{A}(\mathcal{O})$.

10.4 Axiom 4: Poincare Covariance

Let $\rho := (s, a) \in Spin(1,3) \rtimes \mathbb{R}^3 = \mathcal{P}'$ projected to the corresponding element $\tilde{\rho} = (\Lambda, a) \in \mathcal{P}$. Take a representation $\phi : Spin(1,3) \to GL_4(\mathbb{C})$. We define for $f \in C_c^\infty(\mathbb{M}, S), g \in C_c^\infty(\mathbb{M}, S^*)$

$$T_\rho f = \phi(s)(f(\tilde{\rho}^{-1}x)), \quad T_\rho g = g(\tilde{\rho}^{-1}x)\phi(s)^{-1}.$$

The proof is nearly identical to before to show that $\text{supp}(T_\rho f) = \tilde{\rho} \cdot \text{supp}(f)$. We wish to prove the following.

Lemma 10.3. Given $\rho \in \mathcal{P}'$, we have that

$$T_\rho D = D T_\rho, \quad T_\rho D^\dagger = D^\dagger T_\rho, \quad T_\rho S = S T_\rho.$$

It suffices to show that

$$T_\rho \not{\partial} = \not{\partial} T_\rho.$$

Proof. First, note that

$$\partial_\mu T_\rho f(x) = \phi(s)(\Lambda^{-1})^\nu{}_\mu \partial_\nu f(\tilde{\rho}^{-1}x).$$

We need to show that $\phi(s)^{-1}\gamma^\mu\phi(s) = \Lambda^\mu{}_\sigma \gamma^\sigma$. For more discussion on the derivation, see [Ton]. First, we create an antisymmetric tensor encoding the Lorentz boost we use, coming uniquely from Λ. We have generators of the form

$$\Lambda = \exp\left(\frac{1}{2}\Omega_{\rho\sigma}\mathcal{M}^{\rho\sigma}\right) \approx 1 + \frac{1}{2}\Omega_{\rho\sigma}\mathcal{M}^{\rho\sigma}$$

$$\phi(s) = \exp\left(\frac{1}{2}\Omega_{\rho\sigma}S^{\rho\sigma}\right) \approx 1 + \frac{1}{2}\Omega_{\rho\sigma}S^{\rho\sigma}$$

with $\approx$ indicating infinitessimal transformations, and

$$S^{\mu\nu} = \frac{1}{4}[\gamma^\mu, \gamma^\nu], \quad (\mathcal{M}^{\rho\sigma})^\mu_\nu = \eta^{\rho\mu}\delta^\sigma_\nu - \eta^{\sigma\mu}\delta^\rho_\nu.$$

First, we have that

$$[\gamma^\mu\gamma^\nu, \gamma^\rho] = 2(\eta^{\nu\rho}\gamma^\mu - \eta^{\mu\rho}\gamma^\nu)$$

which follows from the relation $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$. Thus, we have that

$$[\eta^{\mu\nu}, \gamma^\rho] = \frac{1}{2}[\gamma^\mu\gamma^\nu, \gamma^\rho] + \frac{1}{2}[\gamma^\nu\gamma^\mu, \gamma^\rho] = 0$$

Now, note that

$$\begin{aligned} S^{\mu\nu} &= \frac{1}{4}\gamma^\mu\gamma^\nu - \frac{1}{4}\gamma^\nu\gamma^\mu \\ &= \frac{1}{2}\gamma^\mu\gamma^\nu - \frac{1}{2}\eta^{\mu\nu} \end{aligned}$$

so we get that

$$\begin{aligned} [S^{\mu\nu}, \gamma^\rho] &= \frac{1}{2}[\gamma^\mu\gamma^\nu, \gamma^\rho] \\ &= \eta^{\nu\rho}\gamma^\mu - \eta^{\nu\rho}\gamma^\nu \end{aligned}$$

but we also have that

$$\begin{aligned} (\mathcal{M}^{\rho\sigma})^\mu{}_\nu\,\gamma^\nu &= \eta^{\rho\mu}\delta^\sigma_\nu\gamma^\nu - \eta^{\sigma\mu}\delta^\rho_\nu\gamma^\nu \\ &= \eta^{\rho\mu}\gamma^\sigma - \eta^{\sigma\mu}\gamma^\rho \\ &= -[S^{\rho\sigma}, \gamma^\mu] \end{aligned}$$

Indeed, we have that

$$\begin{aligned} \phi(s)^{-1}\gamma^\mu\phi(s) &\approx \left(1 - \frac{1}{2}\Omega_{\rho\sigma}S^{\rho\sigma}\right)\gamma^\mu\left(1 + \frac{1}{2}\Omega_{\rho\sigma}S^{\rho\sigma}\right) \\ &= \gamma^\mu - \frac{1}{2}\Omega_{\rho\sigma}[S^{\rho\sigma}, \gamma^\mu] \end{aligned}$$

but we also have that

$$\begin{aligned}\Lambda^{\mu}{}_{\nu}\gamma^{\nu} &\approx \gamma^{\mu} + \frac{1}{2}(\Omega_{\rho\sigma}\mathcal{M}^{\rho\sigma})^{\mu}{}_{\nu}\gamma^{\nu} \\ &= \gamma^{\mu} - \frac{1}{2}\Omega_{\rho\sigma}[S^{\rho\sigma}, \gamma^{\mu}] \\ &\approx \phi(s)^{-1}\gamma^{\mu}\phi(s).\end{aligned}$$

Thus, we have that $\phi(s)^{-1}\gamma^{\mu}\phi(s) = \Lambda^{\mu}{}_{\sigma}\gamma^{\sigma}$, or $\gamma^{\mu}\phi(s) = \phi(s)\Lambda^{\mu}{}_{\sigma}\gamma^{\sigma}$. Putting this together, we have that

$$\begin{aligned}\gamma^{\mu}\partial_{\mu}T_{\rho}f(x) &= \gamma^{\mu}\phi(s)(\Lambda^{-1})^{\nu}{}_{\mu}\partial_{\nu}f(\tilde{\rho}^{-1}x) \\ &= \phi(s)\Lambda^{\mu}{}_{\sigma}\gamma^{\sigma}(\Lambda^{-1})^{\nu}{}_{\mu}\partial_{\nu}f(\tilde{\rho}^{-1}x) \\ &= \phi(s)\delta^{\nu}_{\sigma}\gamma^{\sigma}\partial_{\nu}f(\tilde{\rho}^{-1}x) \\ &= \phi(s)\gamma^{\nu}\partial_{\nu}f(\tilde{\rho}^{-1}x) \\ &= T_{\rho}\not{\partial}f(x).\end{aligned}$$

Thus, since $D = i\not{\partial} - m$, the first equality in the lemma is obvious, similarly for the second because $D^{\dagger} = -i\not{\partial} - m$.

□

This means that we have a well-defined map

$$\alpha(\rho)\psi(f) = \psi(T_{\rho}f), \quad \alpha(\rho)\overline{\psi}(f) = \overline{\psi}(T_{\rho}f)$$

since

$$\begin{aligned}\alpha(\rho)\psi(f + Dg) &= \psi(T_{\rho}f + T_{\rho}Dg) \\ &= \psi(T_{\rho}f + DT_{\rho}g) \\ &= \psi(T_{\rho}f) \\ &= \alpha(\rho)\psi(f)\end{aligned}$$

We also have that for f a spinor and g a cospinor, then

$$\begin{aligned}(T_{\rho}g)(T_{\rho}f)(x) &= g(\tilde{\rho}^{-1}x)\phi(s)^{-1}\phi(s)f(\tilde{\rho}^{-1}x) \\ &= g(\tilde{\rho}^{-1}x)f(\tilde{\rho}^{-1}x)\end{aligned}$$

so this implies that

$$\langle T_{\rho}f, T_{\rho}g\rangle = \langle f, g\rangle$$

since $\det\Lambda = \pm 1$. With $T_{\rho}S = ST_{\rho}$, we also get that

$$\langle T_\rho f, T_\rho S g\rangle = \langle f, Sg\rangle.$$

Thus, this is a $*$-homomorphism on generators. As a result, $\alpha(\rho)$ extends to a $*$-automorphism on $\mathcal{A}(\mathbb{M})$ (**However, this statement along with the version for the Klein-Gordon case have not been rigorously justified yet**).

Remark 10.4. We actually need the domain of α to be an element of $\mathcal{P}$, not $\mathcal{P}'$. To go around this problem, recall that the observables contain an equal number of spinors and cospinors. Thus, regardless of which lift ρ we choose to the spin group for a fixed element $\tilde{\rho} \in \mathcal{P}$, we define a new automorphism $\alpha'(\tilde{\rho})$ such that

$$\alpha'(\tilde{\rho})(\psi(f)\overline{\psi}(g)) = (\alpha(\rho)\psi(f))(\alpha(\rho)\overline{\psi}(g))$$

which is well-defined from the computation of showing $\langle T_\rho f, T_\rho g\rangle = \langle f, g\rangle$.

Lastly, since $\text{supp}(T_\rho f) = \tilde{\rho} \cdot \text{supp}(f)$, $\alpha'(\tilde{\rho})$ is indeed a map such that

$$\alpha'(\tilde{\rho})\mathcal{A}(\mathcal{O}) = \mathcal{A}(\tilde{\rho}\mathcal{O})$$

as desired.

10.5 Axiom 5: Cauchy Surfaces

Given a Cauchy surface $\mathcal{O}_1$ contained in an open set $\mathcal{O}_2$, define $\Sigma_\pm$ and $\chi_\pm$ in a way identical to the case of the Klein-Gordon field. Now, we want to consider for $f \in C_c^\infty(\mathcal{O}_2)$ with

$$k = \chi_+ S^+ f + \chi_- S^- f.$$

For the same reasoning as the Klein-Gordon case, k is compactly supported with $k \in C_c^\infty(\mathcal{O}_2)$. We want to show that for

$$g := f - Dk,$$

then

$$\text{supp}(g) \subseteq \mathcal{O}_1.$$

Similar to before, simplifying gives

$$Dk = f - [D, \chi_+]Sf$$

so that

$$\begin{aligned} g &= [D, \chi_+]Sf \\ &= \chi_+ DSf - D(\chi_+ Sf) \\ &= -D(\chi_+ Sf) \\ &= -i(\not{\partial}\chi_+)Sf - \chi_+(DSf) \\ &= -i(\not{\partial}\chi_+)Sf \end{aligned}$$

which implies that

$$\text{supp}(g) \subseteq \text{supp}(\partial_\mu \chi_+) \cap \text{supp}(Sf) \subseteq \mathcal{O}_1$$

Therefore, we indeed have $\psi(f) = \psi(g)$, and similarly the above computations work for cospinors, just dualize all computations. Thus, since we have shown the needed condition on generators, we have that

$$\mathcal{A}(\mathcal{O}_2) \subseteq \mathcal{A}(\mathcal{O}_1)$$

and with isotony, we get the other direction, proving the time-slice axiom!

11 Future Work, Example 3 - Lattice Gauge Theory and Yang-Mills

The following section involves remarks on future work to be done, and we will utilize our knowledge of lattice gauge theory and Yang-Mills along with AQFT and the Haag-Kastler axioms to discuss modifications to the AQFT framework for a discrete case and apply some reasoning as to how in the continuum limit a Yang-Mills theory should form.

We present a list of axioms that could be considered for a C^*-algebra of observables from lattice gauge theory. Consider a lattice of the form $a\mathbb{Z}^d$.

Axiom 11.1.

1. Local Algebras - For a unital $*$-algebra $\mathcal{A}(\mathbb{M})$ and any open causally convex bounded region $\mathcal{O} \subset \mathbb{M}$, there exists a subalgebra $\mathcal{A}(\mathcal{O})$ containing the unit of $\mathcal{A}(\mathbb{M})$, so that $\{\mathcal{A}(\mathcal{O})\}$ generate $\mathcal{A}(\mathbb{M})$. In particular, this algebra captures observables for the gauge group G, so

$$\mathcal{A}(\mathcal{O})^G = \mathcal{A}(\mathcal{O}).$$

2. Isotony - For $\mathcal{O}_1 \subset \mathcal{O}_2$,

$$\mathcal{A}(\mathcal{O}_1) \subset \mathcal{A}(\mathcal{O}_2).$$

3. Causality - Given two disjoint regions in constant time $\mathcal{O}_1, \mathcal{O}_2$, we have that for all $X \in \mathcal{A}(\mathcal{O}_1)$ and $Y \in \mathcal{A}(\mathcal{O}_2)$

$$[X, Y] = 0.$$

Due to the nature of Hamiltonians on a lattice "spreading" the light cone in a way unlike in the continuum, we seek an estimate bound for regions that are not constant in time. Such a method could be, for example, done through the Lieb-Robinson bound for a finite speed approximation. That is for causally separated regions $\mathcal{O}_1$, and $\mathcal{O}_2$, for all $X \in \mathcal{A}(\mathcal{O}_1)$ and $Y \in \mathcal{A}(\mathcal{O}_2)$, then

$$||[\tau_t^{(a)}(X), Y]|| \leq Ce^{-\mu_a(d(X,Y)-v_a|t|}$$

for $C, \mu_a, v_a > 0$, and

$$\tau_t^{(a)}(X) = e^{itH_a} X e^{-itH_a}.$$

This bound captures an exponential decay of a cone in the lattice, allowing for a certain amount of variance in the causality of spacelike separated observables.

4. Discrete Covariance - Given the symmetry group Γ_a of a cube (or hypercube if a Wick rotation is used and time is discretized!) on $a\mathbb{Z}^d$, we require an automorphism α to exist for each $\rho \in \Gamma_a$ such that

$$\alpha_a(\rho) : \mathcal{A}_a(\mathcal{O}) \to \mathcal{A}_a(\rho\mathcal{O}).$$

5. Existence of Dynamics - Due to the nature of finite speed approximations used for causality, we will use the term **discrete Cauchy lattice**, which is a set N such that given $x \in \mathbb{M}$, we have that any discrete causal curve will intersect N at most once. In particular, the discrete causal curve must be a curve contained within a given cutoff set with respect to the Lieb-Robinson bound. Thus, for $\mathcal{O}_1$ a discrete Cauchy lattice such that $\mathcal{O}_1 \subset \mathcal{O}_2$, then we have that

$$\mathcal{A}(\mathcal{O}_1) = \mathcal{A}(\mathcal{O}_2).$$

Remark 11.2. Similar to the first two examples, an equation of motion coming from the action (in this case, for example, the Wilson action) would prescribe a kernel of solutions to be quotiented by for the domain of functions that observables can take on. The construction, therefore, would rely on algebraic relations and tools like analogs of Green propagators between these observables that permit causality, discrete covariance, and dynamics via discrete Cauchy surfaces.

To describe the algebra of observables on some finite lattice $\Lambda \subset \mathbb{Z}^d$, we need a set of commutation relations. Given unitary operators on a lattice $U(x, y)$ for edge $x, y \in \Lambda$, there are conjugate electric field operators $E_+^a(x, y)$ and $E_-^a(x, y)$ which satisfy

$$[E_+^a(x, y), E_+^b(x, y)] = if^{abc}E_+^c(x, y)$$

$$[E^a_-(x,y), E^b_-(x,y)] = if^{abc}E^c_-(x,y)$$

where f^{abc} are the structure constants for a $G = SU(N)$ [MS15]. Furthermore, because these unitary operators have a gauge transform of the form

$$U(x,y) = g(x)U(x,y)g(y)^{-1}$$

we need to only consider elements that are gauge independent, such as the Wilson loop operator (for neighboring points x_i and x_{i+1} for $i \in \{0, ..., k-1\}$ as well as x_k and x_0.

$$W_l(U) := \mathrm{Tr}(U(x_0,x_1)U(x_1,x_2)...U(x_{k-1},x_k)U(x_k,x_0)).$$

Thus, the GNS construction should yield a state ω_a such that for observables X_a, we have $\omega_a(X_1...X_n)$ as correlator functions on the lattice.

In terms of Yang-Mills theory, there is very little accomplished in the literature, outside of some elementary special cases (e.g. linearized Yang-Mills theory). Nevertheless, we summarize some essential ideas that would be required for a proper Yang-Mills theory. To review, given a compact gauge group G, we have the curvature tensor coming from a gauge field A of the form:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + g[A_\mu, A_\nu], \quad d_A F = 0.$$

What we would seek is the lattice gauge theory's algebra of observables to "converge" in limit as $a \to 0$ to the algebra of observables by Yang-Mills. Thus, we make the following requirement.

Remark 11.3. Let $\mathcal{A}_a(\mathcal{O})$ and $\mathcal{A}_{YM}(\mathcal{O})$ be the algebra of observables supported in $\mathcal{O}$ for a lattice gauge theory on the lattice $a\mathbb{Z}^d$ and for a Yang-Mills theory respectively. Loosely, we require that

$$\lim_{a\to 0} \mathcal{A}_a(\mathcal{O}) = \mathcal{A}_{YM}(\mathcal{O}).$$

More precisely for all observables $\xi_{i,a} \in \mathcal{A}_a(\mathcal{O})$, $\xi_i \in \mathcal{A}_{YM}(\mathcal{O})$ with corresponding representations $\pi_a : \mathcal{A}_a(\mathcal{O}) \to B(\mathcal{H})$ and $\pi_{YM} : \mathcal{A}_a(\mathcal{O}) \to B(\mathcal{H})$ to a common Hilbert space $\mathcal{H}$ (via the GNS construction), the representations should converge in the induced operator norm

$$\lim_{a\to 0} ||\pi_a(\xi_{i,a}) - \pi_{YM}(\xi_i)|| = 0.$$

Isotony will still be maintained trivially. However, causality in the limit would require a nontrivial statement with respect to the Lieb-Robinson bound on finite speed approximation to limit to a true causal disjointness in the continuum and the corresponding adjustment to the Cauchy surfaces would need to hold as well. The symmetry groups of the hypercube would also need to limit to the Poincare group.

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